How language promotes and skews olfactory processing

Norbert Vanek[1,2]
[1]*University of Auckland,*
[2]*Charles University Prague*
norbert.vanek@auckland.ac.nz

Words and scents

BACKGROUND

Language plays a paradoxical role in human olfaction. It can enhance odor categorization while simultaneously impairing recognition, suggesting that verbal influence varies across olfactory processes. Research shows that linguistic richness in smell vocabulary affects odor naming, evaluation, and memory [1,2,3]. While verbal labels can scaffold odor categories, they may also distort recall and hedonic judgments [4]. This duality, where words support cognitive organization yet trigger olfactory illusions [5], aligns with the broader framework of the label-feedback hypothesis [6]. This study explores in three experiments how and in which contexts language facilitates or disrupts olfactory processing.

METHODOLOGY

Exp. 1 Verbal facilitation of odor categories: Native Czech speakers underwent a four-day training to categorize hard-to-identify odors paired either with pseudowords or nonverbal sounds under high (81%) or low (25%) consistency conditions (Fig. 1).

Exp. 2 Limits of verbally assisted odor categorization: Eighty Chinese-English bilinguals were trained over four days to categorize odors using pseudowords in Chinese, English, Georgian, or without labels.

Exp. 3 Verbal skewing of odor memories: 140 participants were exposed to a target odor, completed a language distractor task, and were assigned to one of seven conditions differing in verbal description and response format.

EXPERIMENT 1

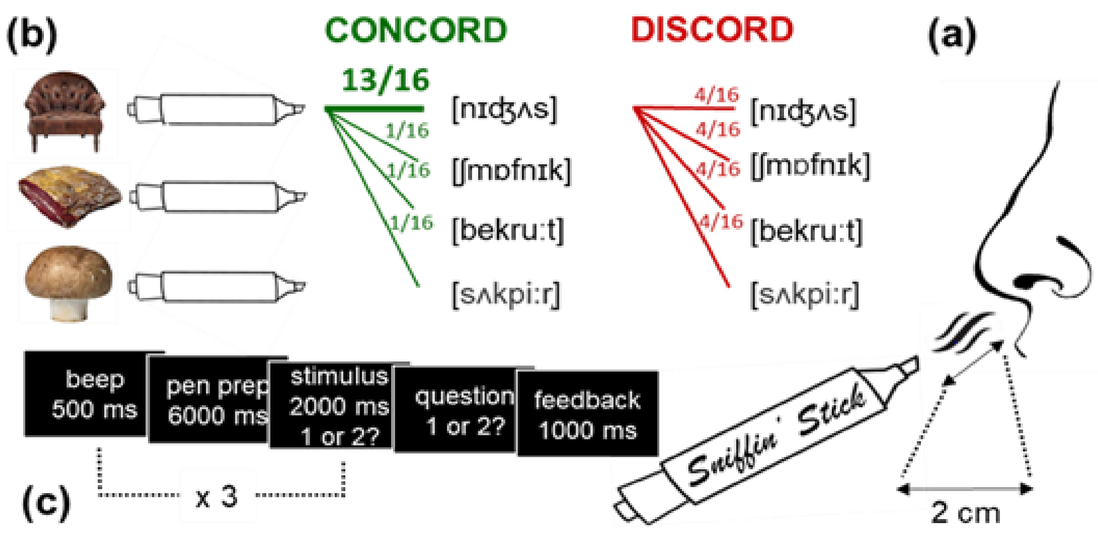

Figure 1. Experiment 1 design using (a) Sniffin' sticks as olfactory stimuli, (b) Bisyllabic pseudowords as verbal stimuli, and (c) odor categorization test without pseudowords

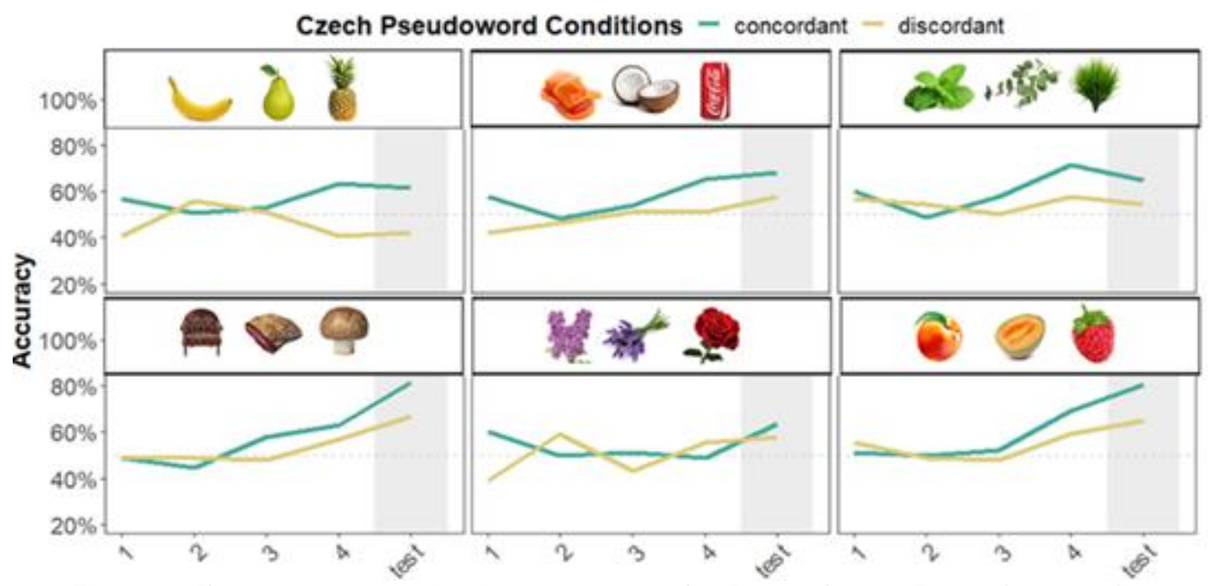

Figure 2. Changes in categorization accuracy in the Concordant vs Discordant conditions per triplet (Exp. 1)

EXPERIMENT 2

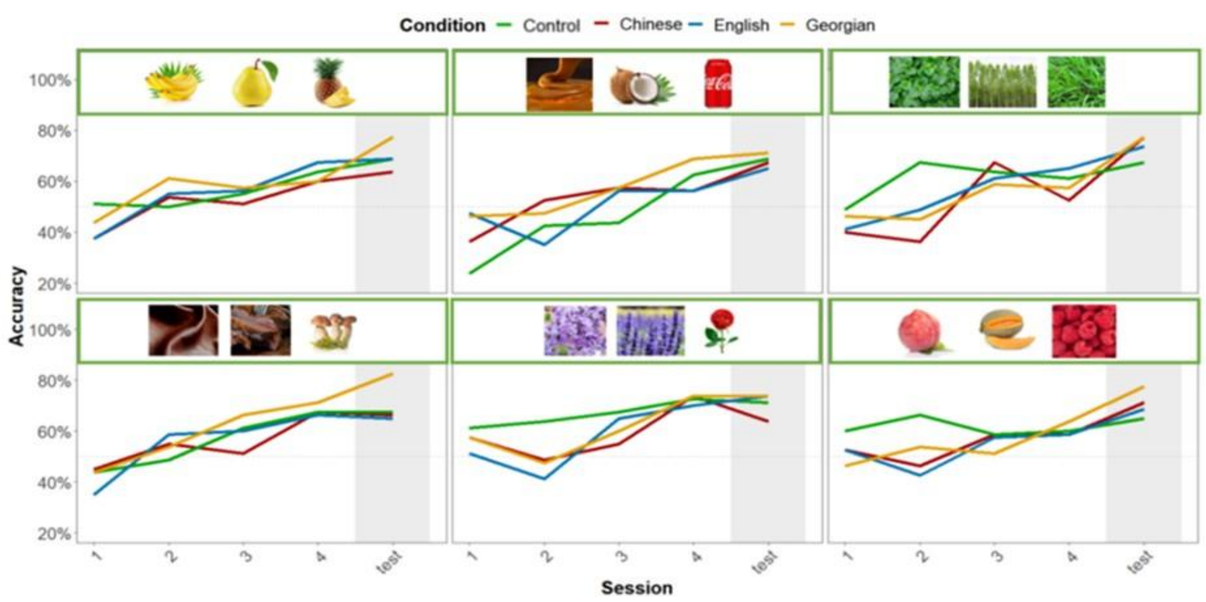

Figure 3. Changes in categorization accuracy over four training sessions and the Test session per triplet (Expt. 2) From: https://doi.org/10.1017/langcog.2025.1

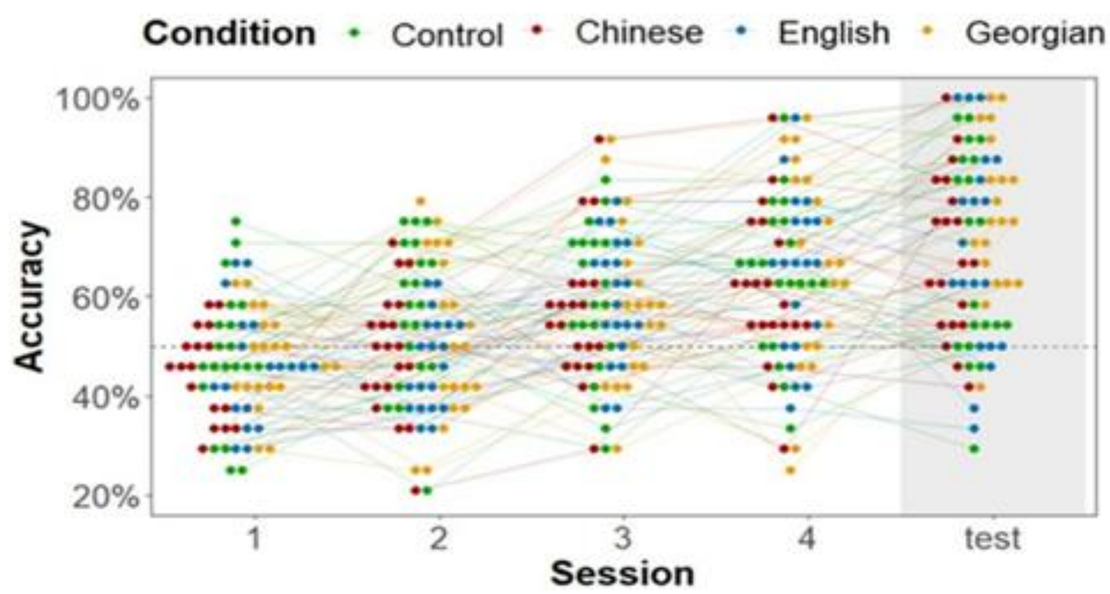

Figure 4. Development of odor categorization accuracy per participant across training sessions and the test (Exp. 2) From: https://doi.org/10.1017/langcog.2025.1

EXPERIMENT 3

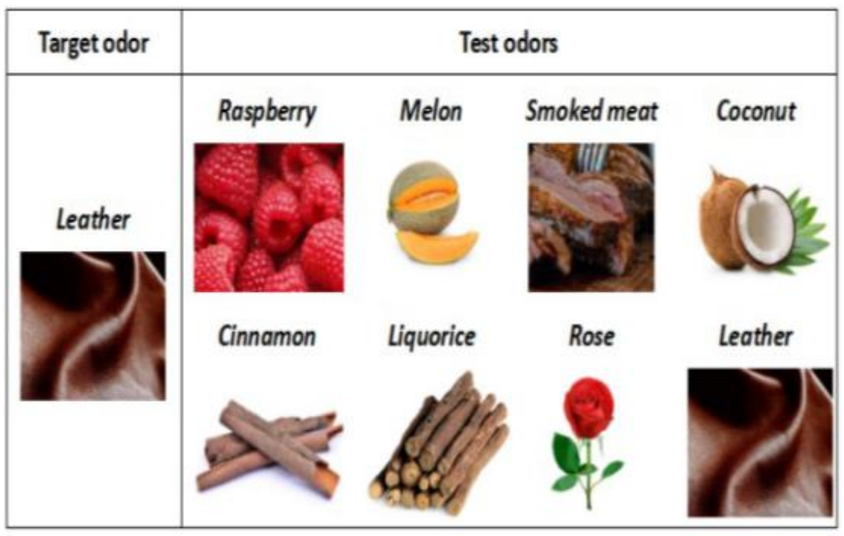

Figure 5. Olfactory stimuli in the verbal overshadowing task (Exp.3) From: Cao, Majid & Vanek (2024) Verbal overshadowing in odor recognition https://escholarship.org/uc/item/46j0b47p

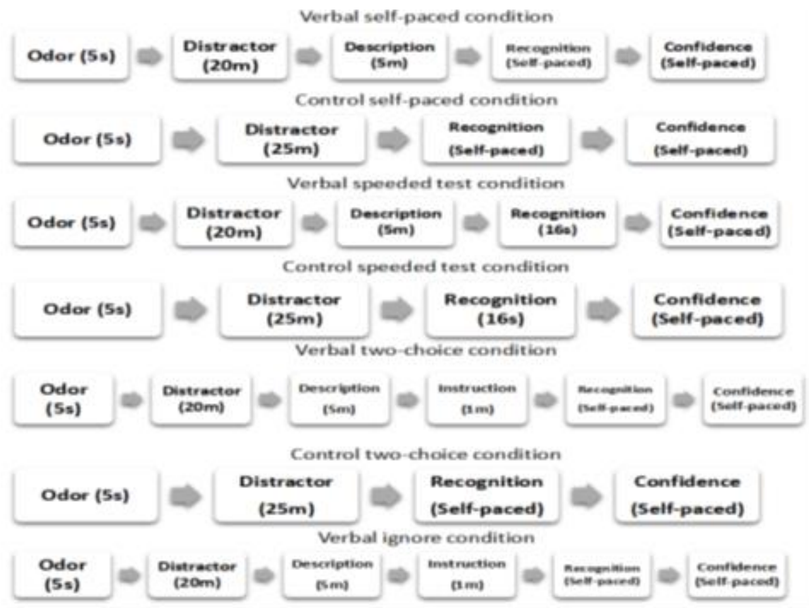

Figure 6. Experimental procedures per condition (Exp. 3) From: Cao, Majid & Vanek (2024) Verbal overshadowing in odor recognition https://escholarship.org/uc/item/46j0b47p

DISCUSSION

This study synthesizes findings from three experiments on olfactory categorization and verbal overshadowing, demonstrating that language plays a dual role in odor cognition, facilitating category learning while potentially distorting recognition. Verbal labels enhance category formation, in line with the label-feedback hypothesis [6], which suggests that labels guide attention toward category-relevant perceptual features. However, Experiment 2 showed that this facilitation is not guaranteed. When label-percept mappings are either inconsistent or cognitively taxing, language may fail to boost learning. In recognition tasks, especially with hard-to-name odors, language can impair accuracy. These results deepen our understanding of how language can both scaffold and skew sensory perception, aligning closely with the 2026 HSP conference theme Language in Minds and Machines by showing how linguistic input dynamically modulates olfactory representations in the human mind.

REFERENCES

- [1] Ayabe-Kanamura, S., Schicker, I., Laska, M., Hudson, R., Distel, H., Kobayakawa, T., & Saito, S. (1998). Differences in perception of everyday odors: a Japanese-German cross cultural study. *Chemical Senses, 23*, 31-38. https://doi.org/10.1093/chemse/23.1.31
- [2] Majid, A., & Burenhult, N. (2014). Odors are expressible in language, as long as you speak the right language. *Cognition, 130(2)*, 266–270.
- [3] Wnuk, E., Laophairoj, R., & Majid, A. (2020). Smell terms are not rara: A semantic investigation of odor vocabulary in Thai. *Linguistics, 58(4)*, 937-966. http://dx.doi.org/10.13039/501100003246
- [4] Herz, R. S. & von Clef, J. (2001). The influence of verbal labeling on the perception of odors: evidence for olfactory illusions? Perception, 30(3), 381-391. https://doi.org/10.1068/p3179
- [5] Herz, R. S. (2005). The unique interaction between language and olfactory perception and cognition. *Trends in Experimental Psychology Research*, 91-109.
- [6] Lupyan, G. (2012). Linguistically modulated perception and cognition: The label-feedback hypothesis. *Frontiers in Psychology, 3*, 54. https://doi.org/10.3389/fpsyg.2012.00054